Chapter 12 Notes

Section 0

Notes

Note that when the relation $x = y$ is used (for instnace in P_1), this is a substitute for the formula

$$\forall z(z \in x \equiv z \in y)$$

Section 1

Exercise 1.1

We show that for finite α, $L_\alpha = R_\alpha$.

Proof. By induction. $L_0 = R_0$ by definition. Suppose $L_n = R_n$ where n is finite. By C_0, L_n is transitive (and thus also extensional), and hence by P_3, $\mathcal{F}(L_n) = \mathcal{P}(L_n) = \mathcal{P}(R_n) = R_{n+1}$, therefore completing the induction. □

Exercise 1.2

By Exercise 1.1, for finite α, $L_\alpha = R_\alpha$. Then it obviously follows that $L_\omega = R_\omega$. However, $L_{\omega+1} \neq R_{\omega+1}$. For recall that, if a set A is infinite (and has cardinality), then $\mathcal{F}(A) \preceq A$. Thus $L_{\omega+1} = \mathcal{F}(L_\omega) \preceq L_\omega$ (because L_ω is infinite). But $R_{\omega+1} = \mathcal{P}(R_\omega) \succ R_\omega$ ($\succ L_\omega$) by Cantor's Theorem. Hence $L_{\omega+1} \neq R_{\omega+1}$, and $R_{\omega+1} \succ L_{\omega+1}$.

The fact that A has cardinality in the above argument follows for this reason: Obviously each L_n for a natural number n has cardinality (because they are all finite). Then L_ω is denumerable (hence has cardinality), and by the argument given in the notes of last chapter, it then also follows that $L_{\omega+1}$ is denumerable.

Section 2

Notes

It seems that ϕ being absolute over K is in fact equivalent to saying that K reflects V with respect to ϕ.

The definition given here of the absoluteness of a class over K is clearly not correct. For, it says that: "We shall say that a class is absolute over K if it is defined over K by at least one formula that is absolute over K." But, under this definition, it would then follow that the only absolute class is $\emptyset$! For suppose we had an absolute class A. Then, for every transitive class K, A would be definable *over* K. But $\emptyset$ is transitive. Thus A would be definable over $\emptyset$, meaning $A = \emptyset$.

It seems clear to me that the definition of a class being absolute should be changed to: "We shall say that a class is absolute over K if it is defined *over* V by at least one formula that is absolute over K." This seems to be how the book is using the notion of absoluteness. Taking this definition also solves a problem with Theorem 2.2. It is claimed that this proves that every Σ_0-class and relation is absolute; however, the proof only handles the case of formulas. With the previous definition, we certainly would be unable to prove the desired result. But, with the altered definition, consider any Σ_0 class A. Thus it is definable over V by some Σ_0 formula $\phi(x)$. But we have already shown that $\phi(x)$ is absolute (over any transitive class K). Thence it follows that A is absolute.

Note the following fallacious argument: "In fact, after showing that $x \in y$ is absolute, it immediately follows that $x \in y^+$ is absolute, as it is a special case of $x \in y$ being absolute." This fails because y^+ is not a variable.

It is remarked in (15) that "We already know that the condition 'x is transitive' is absolute over any transitive class". I do not know where we established this before this exercise.

The formula provided in (26) does not seem correct. For, in fact, under this definition, it would follow that all ordered pairs were unordered pairs. Because, for any ordered pair $x : \langle y, z\rangle = \{\{y\}, \{y, z\}\}$, there are some w and v such that $x = \{w, v\}$, namely when $w = \{y\}$ and $v = \{y, z\}$ (or the other way around). A better definition would be, intuitively, x is an ordered pair iff there are no elements y, z such that $x = \langle y, z\rangle$, or, formally:

$$(\forall y \in x)(\forall z_1 \in y)(\forall z_2 \in y)(x \neq \langle z_1, z_2\rangle)$$

The formula given for (29) does not seem correct to me. It needs to show that there are no $\langle y, z_1\rangle \in x$ and $\langle y, z_2\rangle \in x$ where $z_1 \neq z_2$, yet it does not guarantee this. Instead, we can define (29) using (30) and (31). x is a function iff

$$x \text{ is a relation} \wedge (\forall y \in \mathrm{Dom}(x))(\forall z_1 \in \mathrm{Ran}(x))(\forall z_2 \in \mathrm{Ran}(x))$$

$$((\langle y, z_1\rangle \in x \wedge \langle y, z_2\rangle \in x) \supset z_1 = z_2)$$

Here is the proof of (31). First, note that $y \in \mathrm{Ran}(x)$ is Σ_0, as $y \in \mathrm{Ran}(x)$ iff $(\exists z \in x)(\exists w \in z)(\exists v \in w)(z = \langle v, y\rangle)$. Then $y = \mathrm{Ran}(x)$ iff

$$(\forall z \in y)(z \in \mathrm{Ran}(x)) \wedge (\forall z_1 \in x)(\forall z_2 \in z_1)(\forall z_3 \in z_2)$$

$$(z_3 \in \mathrm{Dom}(x) \subset z_3 \in y)$$

Section 3

Lemma 3.5*

The proof of this is not correct. In particular, it is claimed that $b = R''(a)$. However, because R is only a relation, it could be that, for some $x \in a$, xRy for some $y \notin b$, which means it is possible that $b \neq R''(a)$. This is, however, not an issue, as obviously $b \subseteq R''(a)$ (by definition of a).

Lemma 3.6

The proof of this Lemma requires a correction. It is claimed that, because the relation $\mathcal{Q}(x) = y$ is absolute, it follows that it is defined by some absolute formula $\phi(x, y)$. However, this was not the definition of a relation being absolute. Instead, the definition said that it is absolute over every transitive class K by *some* formula ϕ (which is absolute over K). This does not seem to necessarily imply that the relation is defined by $\phi(x, y)$ *simultaneously* over every transitive class K. However, consider L_α in the proof of this Lemma. We know that it is transitive, so there is indeed some $\phi(x, y)$ that is absolute over L_α that defines the relation.

Then clearly the formula

$$(1)\ \exists y(y \in c \wedge \phi(y, x))$$

defines $\mathcal{Q}''(c)$ over L_α, as all constants in (1) are in L_α. Also $\phi(y, x)$ is absolute over L_α, meaning (1) is true over L_α iff $\phi(y, x)$ is true over L_α for some $y \in c$ iff $\phi(y, x)$ is true over V for some $y \in c$ iff $x \in \mathcal{Q}''(x)$.

Proposition 3.7

There is an error in the proof of this proposition. The Σ_0 formula said to define the relation $\mathcal{Q}(x) = y$ actually defines the more general relation: $\mathcal{Q}(\langle a, b\rangle) = a$, but we do not necessarily need to have that a, b be constructible sets (as the definition in the book is given). This is not an issue, as this altered relation still works with Lemma 3.6: In fact, even if $\mathcal{Q}$ is not a function from constructible sets to constructible sets, the proof of Lemma 3.6 still works (or if it is a functions whose domain or range is not just L).

Proposition 3.8

It seems to me that $L^n_{\overline{a},i}$ is also a constructible class, as we can just replace the part in the proof saying $x_i \in x_j$ by $a \in x_i$. Similarly, to show that $L^n_{\overline{a},\overline{b}}$ is a constructible class, we can replace the part in the proof saying $x_i \in x_j$ by $a \in b$.

Exercise 3.1

(a) We show that $L^n_{\overline{a},i}$ is a constructible class, where a is a constructible set. Now, $L^n_{\overline{a},i} \subseteq L^n$, which of course means $L^n_{\overline{a},i}$ is a class of constructible sets. Thus we must show that $L^n_{\overline{a},i} \cap c$ is a constructible set, where c is any constructible set. So let c be any constructible set. Let α be an ordinal such that $c \in L_\alpha$ and $a \in L_\alpha$ (which exists because a is a constructible set). Then the formula

$$x \in c \wedge (\exists x_1 \in c)(\exists x_2 \in c)...(\exists x_n \in c)(x = \langle x_1, ..., x_n\rangle \wedge a \in x_i)$$

defines $L^n_{\overline{a},i}$ over L_α (because both a and c are in L_α).

(b) We show that $L^n_{\overline{a},\overline{b}}$ is a constructible class, where a and b are constructible sets. Now, $L^n_{\overline{a},\overline{b}}$ is either L^n or $\emptyset$. But by Proposition 3.8, L^n is a constructible class; hence $L^n_{\overline{a},\overline{b}}$ is a constructible class (because $\emptyset$ is of course also a constructible class).